

VISVESVARAYA TECHNOLOGICAL UNIVERSITY, BELAGAVI
B.E. in Electronics Engineering (VLSI Design & Technology)
Scheme of Teaching and Examinations 2022
 Outcome Based Education (OBE) and Choice Based Credit System (CBCS)
 (Effective from the academic year 2023-24)

III SEMESTER

Sl. No.	Course	Course Code	Course Title	Teaching Department (TD) and Question Paper Setting Board (PSB)	Teaching Hours /Week				Examination				Credits
					Theory Lecture	Tutorial	Practical / Discussion	SDA	Duration in hours	CIE Marks	SEE Marks	Total Marks	
					L	T	P	S					
1	PCC	BEC301		TD- Maths PSB - Maths	3	0	0		03	50	50	100	3
2	IPCC	BVL302	Analog Electronic Circuits	TD: ECE / EE(VDT) PSB: ECE/ EE(VDT)	3	0	2		03	50	50	100	4
3	IPCC	BVL303	Digital Logic Circuits	TD: ECE/ EE(VDT) PSB: ECE/ EE(VDT)	3	0	2		03	50	50	100	4
4	PCC	BVL304	Verilog HDL	TD: ECE/ EE(VDT) PSB: ECE/ EE(VDT)	3	0	0		03	50	50	100	3
5	PCCL	BVLL305	Verilog HDL LAB	TD: ECE/ EE(VDT) PSB: ECE/ EE(VDT)	0	0	2		03	50	50	100	1
6	ESC	BXX306x	ESC / ETC / PLC	TD: PSB:	3	0	0		03	50	50	100	3
7	UHV	BSCK307	Social Connect and Responsibility	Any Department	0	0	2		01	100	---	100	1
8	AEC / SEC	BVL358x	Ability Enhancement Course / Skill Enhancement Course - III		If the course is a Theory				01	50	50	100	1
					1	0	0						
					If a course is a laboratory				02				
					0	0	2						
9	MC	BNSK359	National Service Scheme (NSS)	NSS coordinator	0	0	2			100	---	100	0
		BPEK359	Physical Education (PE) (Sports and Athletics)	Physical Education Director									
		BYOK359	Yoga	Yoga Teacher									
Total										550	350	900	20

PCC: Professional Core Course, **PCCL:** Professional Core Course laboratory, **UHV:** Universal Human Value Course, **MC:** Mandatory Course (Non-credit), **AEC:** Ability Enhancement Course, **SEC:** Skill Enhancement Course, **L:** Lecture, **T:** Tutorial, **P:** Practical **S= SDA:** Skill Development Activity, **CIE:** Continuous Internal Evaluation, **SEE:** Semester End Evaluation. K : This letter in the course code indicates common to all the stream of engineering. ESC: Engineering Science Course, ETC: Emerging Technology Course, PLC: Programming Language Course

Engineering Science Course (ESC/ETC/PLC)			
BVL306A	Circuits & Controls	BVL306C	Computer Organization and Architecture
BVL306B	Sensors and Instrumentation	BVL306D	Physics of semiconductor devices
Ability Enhancement Course – III			
BVLL358A	UNIX Programming	BVL358C	Digital Engineering Course (NASSCOM)
BVLL358B	Circuit Laboratory using P-spice	BVLL358D	IoT (Internet of Things) Lab
Professional Core Course (IPCC): Refers to Professional Core Course Theory Integrated with practicals of the same course. Credit for IPCC can be 04 and its Teaching– Learning hours (L : T : P) can be considered as (3 : 0 : 2) or (2 : 2 : 2). The theory part of the IPCC shall be evaluated both by CIE and SEE. The practical part shall be evaluated by only CIE (no SEE). However, questions from the practical part of IPCC shall be included in the SEE question paper. For more details, the regulation governing the Degree of Bachelor of Engineering /Technology (B.E./B.Tech.) 2022-23 may please be referred. National Service Scheme /Physical Education/Yoga: All students have to register for any one of the courses namely National Service Scheme (NSS), Physical Education (PE)(Sports and Athletics), and Yoga(YOG) with the concerned coordinator of the course during the first week of III semesters. Activities shall be carried out between III semester to the VI semester (for 4 semesters). Successful completion of the registered course and requisite CIE score is mandatory for the award of the degree. The events shall be appropriately scheduled by the colleges and the same shall be reflected in the calendar prepared for the NSS, PE, and Yoga activities. These courses shall not be considered for vertical progression as well as for the calculation of SGPA and CGPA, but completion of the course is mandatory for the award of degree.			

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IV SEMESTER													
Sl. No	Course and Course Code		Course Title	Teaching Department (TD) and Question and Paper Setting Board (PSB)	Teaching Hours /Week				Examination				Credits
					Theory Lecture	Tutorial	Practical/ Drawing	Self-Study	Duration in hours	CIE Marks	SEE Marks	Total Marks	
1	PCC	BVL401	FPGA Based System design Using Verilog	TD: ECE /VDT PSB: ECE/VDT	3	0	0		03	50	50	100	3
2	IPCC	BVL402	Principles of Communication Systems	TD: ECE /VDT PSB: ECE/VDT	3	0	2		03	50	50	100	4
3	IPCC	BVL403	Control Systems	TD: ECE /VDT PSB: ECE/VDT	3	0	2		03	50	50	100	4
4	PCCL	BVLL404	FPGA Based System design Using Verilog Lab	TD: ECE /VDT PSB: ECE/VDT	0	0	2		03	50	50	100	1
5	ESC	BVL405x	ESC/ETC/PLC	TD: ECE /VDT PSB: ECE/VDT	3	0	0		03	50	50	100	3
6	AEC/ SEC	BXX456x	Ability Enhancement Course/Skill Enhancement Course- IV	TD and PSB: Concerned department	If the course is Theory				01	50	50	100	1
					1	0	0						
					If the course is a lab				02				
					0	0	2						
4	BSC	BBOK407	Biology For Engineers	TD / PSB: BT, CHE,	3	0	0		03	50	50	100	3
7	UHV	BUHK408	Universal human values course	Any Department	1	0	0		01	50	50	100	1
9	MC	BNSK459	National Service Scheme (NSS)	NSS coordinator	0	0	2			100	---	100	0
		BPEK459	Physical Education (PE) (Sports and Athletics)	Physical Education Director									
		BYOK459	Yoga	Yoga Teacher									
Total										500	400	900	20

PCC: Professional Core Course, **PCCL:** Professional Core Course laboratory, **UHV:** Universal Human Value Course, **MC:** Mandatory Course (Non-credit), **AEC:** Ability Enhancement Course, **SEC:** Skill Enhancement Course, **L:** Lecture, **T:** Tutorial, **P:** Practical **S= SDA:** Skill Development Activity, **CIE:** Continuous Internal Evaluation, **SEE:** Semester End Evaluation. **K:** This letter in the course code indicates common to all the stream of engineering.

Engineering Science Course (ESC/ETC/PLC)			
BVL405A	Microcontrollers	BVL405C	Operating Systems
BVL405B	Industrial Electronics	BVL405D	Data Structures using C
Ability Enhancement Course / Skill Enhancement Course - IV			
BVL456A	Microcontrollers Lab	BVL456C	Octave Programming
BVL456B	Programmable Logic Controllers	BVL456D	Data Structures Lab using C
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